UNIVERSITY OF LONDON

CM3060

BSc EXAMINATION

COMPUTER SCIENCE

Natural Language Processing

Release date: Tuesday 6 September 2022 at 12:00 midday British Summer Time

Submission date: Wednesday 7 September 2022 by 12:00 midday British Summer Time

Time allowed: 24 hours to submit

INSTRUCTIONS TO CANDIDATES:

Section A of this assessment paper consists of a set of **TEN** Multiple Choice Questions (MCQs) which you will take separately from this paper. You should attempt to answer **ALL** the questions in Section A. The maximum mark for Section A is **40**.

Section A will be completed online on the VLE. You may choose to access the MCQs at any time following the release of the paper, but once you have accessed the MCQs you must submit your answers before the deadline or within **4 hours** of starting whichever occurs first.

Section B of this assessment paper is an online assessment to be completed within the same 24-hour window as Section A. We anticipate that approximately **1 hour** is sufficient for you to answer Section B. Candidates must answer **TWO** out of the THREE questions in Section B. The maximum mark for Section B is **60**.

Calculators are not permitted in this examination. Credit will only be given if all workings are shown.

You should complete **Section B** of this paper and submit your answers as **one document**, if possible, in Microsoft Word or a PDF to the appropriate area on the VLE. Each file uploaded must be accompanied by a coversheet containing your **candidate number**. In addition, your answers must have your candidate number written clearly at the top of the page before you upload your work. Do not write your name anywhere in your answers.

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SECTION A

Candidates should answer the **TEN** Multiple Choice Questions (MCQs) quiz, **Question 1** in Section A on the VLE.

SECTION B

Candidates should answer any **TWO** questions from Section B.

Question 2

In a 100-million-word corpus we find the following counts of unigrams and bigrams:

my	883,614
beliefs	80,891
chimneys	21
my beliefs	378
my chimneys	0

(a) Estimate the probabilities P(beliefs) and P(beliefs | my) using Maximum Likelihood estimation (without smoothing). Express your answer as a fraction.

[5]

(b) Estimate the bigram probability P(chimneys | my) using Maximum Likelihood Estimation and Laplace smoothing. Assume that the vocabulary consists of 1,234,567 unique words. Express your answer as a fraction.

[5]

(c) Estimate the bigram probability P(chimneys | my) using Maximum Likelihood Estimation and add-k smoothing with k = 0.1. Assume that the vocabulary consists of 1,234,567 unique words. Express your answer as a fraction.

[5]

(d) What other types of smoothing could you consider? Compare and contrast alternative methods with those above

[5]

(e) Suppose that you train three n-gram models using 50 million words from Wikipedia: a unigram model, a bigram model, and a trigram model. Your colleague evaluates the models on 5 million words of test data and obtains the following perplexity scores: 234, 123, and 876 respectively. Does this sound right to you? If not, why not?

[5]

(f) Mark the noun groups in the following sentence using BIO (beginning, intermediate, other) tags:

[5]

Noshin has a car with a sunroof and a set of alloy wheels. It also has go-faster stripes and a pair of furry dice.

Question 3

CC Coordinating conjunction	RB Adverb
CD Cardinal number	RBR Adverb, comparative
DT Determiner	RBS Adverb, superlative
EX Existential there	RP Particle
FW Foreign word	SYM Symbol
IN Preposition or subordinating conjunction	TO to
JJ Adjective	UH Interjection
JJR Adjective, comparative	VB Verb, base form
JJS Adjective, superlative	VBD Verb, past tense
LS List item marker	VBG Verb, gerund or present participle
MD Modal	VBN Verb, past participle
NN Noun, singular or mass	VBP Verb, non-3rd person singular present
NNS Noun, plural	VBZ Verb, 3rd person singular present
NNP Proper noun, singular	WDT Wh-determiner
NNPS Proper noun, plural	WP Wh-pronoun
PDT Predeterminer	WP$ Possessive wh-pronoun
POS Possessive ending	WRB Wh-adverb
PRP Personal pronoun	PU Punctuation
PRP$ Possessive pronoun	

Table 1: Treebank POS tags

(a) Assign parts of speech tags (as per Table 1) to all the words in the following two sentences using the notation word/POS.

[6]

Rick and Francis wrote a song with three chords.

It was played in a very large hall near their studio.

(b) Draw a phrase structure diagram representing one parse of the following sentence. Make a list of the phrase structure rules that you are assuming.

[10]

Francis and Rick wrote a song with three chords

(c) Given the following training data, calculate the likelihood probabilities for each word given each POS.

[6]

- house/NNS moving/VBG is/VBZ stressful/JJ
- moving/JJ houses/NNS are/VBZ dangerous/JJ
- I/PRP saw/VBZ Essi/NNP moving/VBG houses/NNS
- She/PRP houses/VBZ refugees/NNS

(d) Draw a finite state machine where states are parts of speech, and the edges are labelled with transition probabilities.

[8]

Question 4

(a) Write regular expressions that recognize the following:

(i) any string that contains at least four digits

[2]

(ii) any string that starts with one uppercase character, and either ends with 3 digits or with 2 vowels

[4]

(b) Briefly define the morphology operations: compounding, derivation, inflection. Given the root ‘rain’ give an example of the result of applying each operation.

[6]

(c) What is the distributional hypothesis and how is it related to word embeddings?

[6]

(d) You are given the task of designing a classifier to perform sentiment analysis on product reviews. Here are two sample sentences from such reviews:

- The 950cc engine of the Toyota Aygo seemed sluggish and took for ever to reach 62kmph but the Fiat 500 with a similar capacity motor felt very peppy and beat the Aygo hands-down in the 62kmph race.

- The Renault has a taut, athletic look with flowing lines which gives it a solid yet fresh look. In contrast the Dacia looks like a box with flat panels and harsh edges.

What are the main tasks in extracting sentiment from data such as this? You do not have to give a detailed architecture, just list the main challenges.

[6]

(e) What are the values for accuracy and F-measure in the following confusion matrix? [6]

	Predicted positive	Predicted negative
True positives	10	20
True negatives	30	40

END OF PAPER